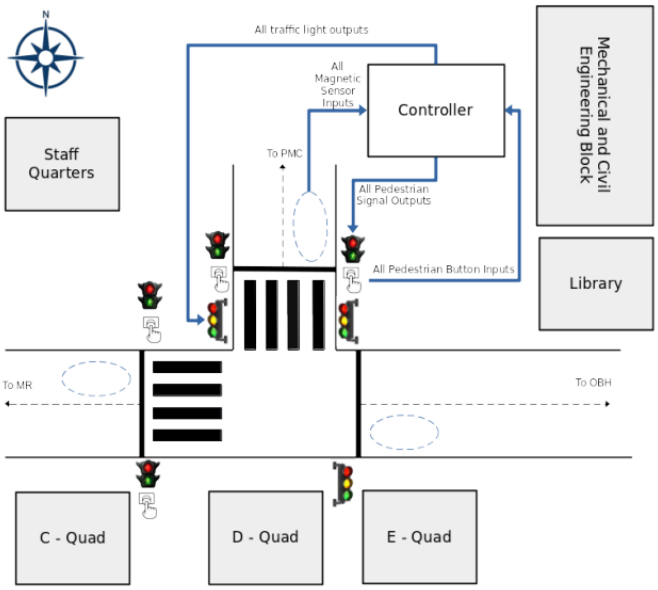
Assignment 1



The assignment has been divided into 3 different stages as there are three 'intelligent' behaviour and one normal.

1. In this we design a normal traffic system in which the only vehicles traffic is considered with no intelligent behaviour.
2. Here we take the First design and add the pedestrian sequence in it to achieve first intelligent behaviour.

3. In last stage we add the two intelligent behaviour related to the i/p of the magnetic sensors with the system last upgraded.

1. For Only Traffic Light System (vehicles)

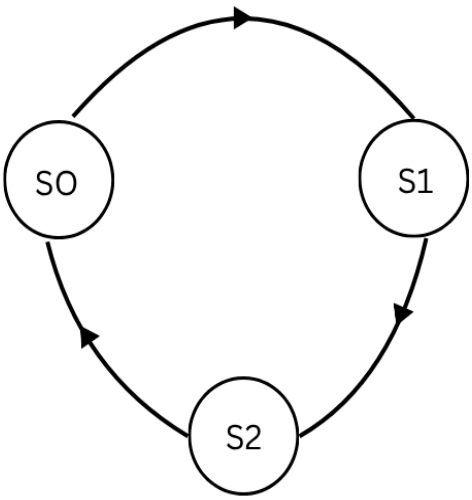
Assuming (East - West) = S0,

(MR - PMC) = S1,

(OBH-PMC) = S2,

Green = G, Yellow = Y and Red = R.

Then the state diagram will be:



For S0 (allowing East – West traffic)

G = 10 sec

Y = 4 sec

All R = 2 sec

For S1 (allowing MR – PMC traffic)

G = 10 sec

Y = 4 sec

All R = 2 sec

For S2 (allowing OBH – PMC traffic)

G = 10 sec

Y = 4 sec

All R = 2 sec

- Summarizing everything in table we get;

Without directional arrows

OBH	PMC	MR	O/P
1	0	1	S0
0	1	1	S1
1	1	0	S2

With directional arrows

OBH		PMC		MR		O/P
(↑)	(→)	(←)	(→)	(↑)	(←)	
1	0	0	0	1	0	S0
0	0	0	1	0	1	S1
0	1	1	0	0	0	S2

2. Pedestrian Crossing sequence (added to vehicle traffic)

The pedestrian crossing state is considered as (**S3)**

- Once the pedestrian button is pressed all the traffic light goes Red for Vehicles

Then for the pedestrian the light is shown as follows;

Green = 10 sec

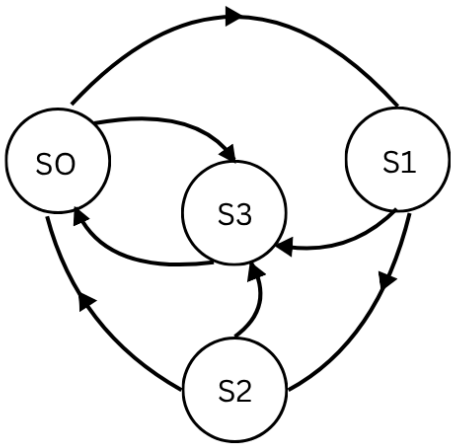
Yellow = 6 sec

All Red = 2 sec

- Assuming the state of button being pressed as **(B)** , button not being pressed as **(B')** as 0's and 1's.

Button's State	Output
0	(B')
1	(B)

Now the state diagram along with the pedestrian crossing state will be



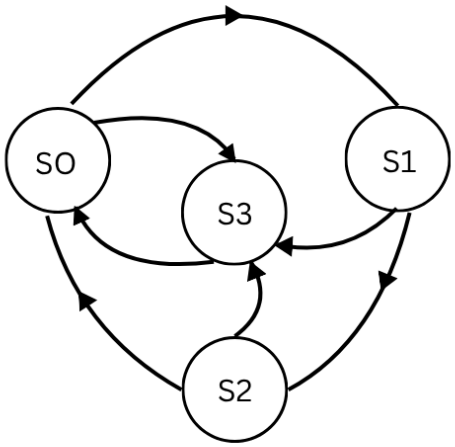
In the above state diagram the normal traffic states will be continued until the input from the button become high (**B**). Once the i/p is high the current sequence of the vehicals traffic has stoped and the pedesterian sequence begin. After that complition of the pedestrian sequence the vehicles traffic sequence start from S0 and go on until the next interupt.

Once the pedestrian button is pressed all other input to the pedestrian buttons are ignored during the sequence.

Summarizing all the inputs and states into a table we get(B = i/p from the pedestrian button)

B	S0	S1	S2	O/P
0	1	0	0	OBH - MR
0	0	1	0	MR - PMC
0	0	0	1	OBH - PMC
1	X	X	X	Pedestrian (S3)

3 Adding the Second and third intelligent behaviour related to the magnetic sensors.



The same state diagram is used with the addition of the i/p coming from the magnetic sensors .

In this state diagram the state chages occurs as following :-

1. From S0 → S1 or S1 → S2 or S2 → S0 only if = (V and B’) occurs

2. From S0 → S3 or S1 → S3 or S2 → S3 only if = (V’ or B) occurs

3. From S3 → S0 occurs if = (V or B’)

Where the i/p are

:- when there is even one incoming vehicle = 1.

:- when there no vehicle = 0.

Magnetic sensor i/p	o/p
0	V’
1	V

The relation of the pedestrian and the magnetic sensor is shown below and o/p are states

V	B	o/p
0	X	S3
X	1	S3

The below table shiws the relation between the current state and next-state with resspect to the i/p from ‘V’ and ‘B’ respectively.

Current state	I/P		Next-state
S	V	B	S’
S0	0	X	S3
S0	1	0	S1
S0	X	1	S3
S1	0	X	S3
S1	1	0	S2
S1	X	1	S3
S2	0	X	S3
S2	1	0	S0
S2	X	1	S3

And if we take the states as the binary then :-

states	encoding
S0	00
S1	01
S2	10
S3	11

After comhining the the encoded state to the state tanle we get :-

Current state		I/P		Next-state	
S_0	S_1	V	B	S'_0	S'_1
0	0	0	X	1	1
0	0	1	0	0	1
0	0	X	1	1	1
0	1	0	X	1	1
0	1	1	0	1	0
0	1	X	1	1	1
1	0	0	X	1	1
1	0	1	0	0	0
1	0	X	1	1	1